Team 32-603 - GFP Design Rationale

2026 Protein Design in SynBio Challenges

Team Strategy Overview

Design Philosophy: Data-driven computational optimization + Literature-validated mutations

Four complementary approaches across 5 different GFP scaffolds:

A. FoldX-driven stabilization -- Identify mutations with negative DeltaDeltaG via FoldX BuildModel (G1, G5)

B. Brightness optimization on naturally stable scaffolds -- Apply brightness data to ultra-stable WT scaffolds (G6, G8)

C. Cross-species superfolder transplant -- Import proven folding mutations between species (G10)

D. Literature-verified single mutation -- Use experimentally validated mutation from published work (G12)

Design 1 - sfGFP_Plus_v1 (32-603_1)

Scaffold: sfGFP (PDB: 2B3P, 238 aa) | WT DeltaG: +3.40 kcal/mol | Strategy: FoldX-driven stabilization

Mutations: Q69L (usGFP core packing), S147P (FoldX DDG=-2.53, loop rigidity), N149K (FoldX DDG=-3.14), N164Y (usGFP core), E172K (FoldX DDG=-4.01, best single mutation), D190N (FoldX DDG=-2.66), A227V (FoldX DDG=-1.91)

Logic: Take competition baseline sfGFP and fix its main weakness -- marginal stability (WT DeltaG=+3.40). E172K alone brings DeltaG to -0.61. Stacking 7 independent stabilizing mutations yields a highly stable variant while maintaining brightness.

Design 2 - amacGFP_Plus_v3_max (32-603_2)

Scaffold: amacGFP (PDB: 7LG4, 238 aa) | WT DeltaG: +3.55 | WT Brightness: 3.97

Strategy: Maximum safe FoldX optimization

8 mutations all with DDG < -2.5: Y39N (-4.36), Y151F, S147P (-2.53), D190N (-2.66), N149K (-4.06), E172K (-3.06), T63A (-2.79), I171V (-3.18)

Logic: amacGFP is naturally bright (1.8x avGFP) but unstable. FoldX shows 8/11 tested mutations are strongly stabilizing. We incorporate all 8 for maximum stability while preserving brightness.

Design 3 - ppluGFP_Opt_v1_bright (32-603_3)

Scaffold: ppluGFP (PDB: 2G6X, 222 aa) | WT DeltaG: -49.88 (ultra-stable) | WT Brightness: 4.23

Strategy: Brightness optimization on naturally ultra-stable scaffold

Mutations: K48E, S118E (TGP surface negative charge), L199H, G206E, A221E (brightness optimization)

Logic: ppluGFP is naturally 25x more stable than sfGFP. Focus entirely on brightness. L199H, G206E, A221E from brightness data; K48E, S118E add negative surface charges to prevent aggregation.

Design 4 - cgreGFP_Opt_v1_bright (32-603_4)

Scaffold: cgreGFP (PDB: 2HPW, 235 aa) | WT DeltaG: -26.89 | WT Brightness: 4.50 (6x avGFP)

Strategy: Minimal modification on brightest natural scaffold

Mutations: K167M (brightest mutant, 4.57), A168V (4.55), M172E (4.56), R78G (loop flexibility)

Logic: cgreGFP is the brightest WT GFP known (~6x avGFP). Only 4 brightness-optimizing mutations to push further without compromising exceptional stability.

Design 5 - avGFP_Superfold (32-603_5)

Scaffold: avGFP (PDB: 2WUR, 238 aa) | WT DeltaG: -2.02 | WT Brightness: 3.72

Strategy: Import sfGFP superfolder mutations into avGFP

Key mutations: S30R, V163A (superfolder keys), Y39N, N149K, E172K, M153T, I171V, S147P, D190N, A227V, Q80R

Logic: Systematically import sfGFP superfolder mutations (Pedelacq 2006) into avGFP backbone combined with FoldX-validated stabilizers. Creates hybrid with avGFP folding + sfGFP stability.

Design 6 - sfGFP_N39D_Frenzel (32-603_6)

Scaffold: sfGFP (PDB: 2B3P, 238 aa) | Strategy: Literature-verified mutation

Key mutation: N39D -- Frenzel 2018 demonstrated 885-fold fluorescence increase at 60C

Superfolder framework: Q80R, S147P, N149K, M153T, V163A, I171V, E172K, D190N, A227V

Logic: Frenzel et al. (2018) proved single N39D mutation causes 885-fold fluorescence increase at 60C via improved folding. Combined with established superfolder framework. Serves as literature-benchmarked validation.

Summary

Seq_ID	Design	Scaffold	Len	Strategy
32-603_1	sfGFP_Plus_v1	sfGFP	238	FoldX stabilization
32-603_2	amacGFP_v3_max	amacGFP	238	Max FoldX opt
32-603_3	ppluGFP_v1	ppluGFP	222	Ultra-stable+bright
32-603_4	cgreGFP_v1	cgreGFP	235	Brightest+bright opt
32-603_5	avGFP_Superfold	avGFP	238	Superfolder transplant
32-603_6	sfGFP_N39D	sfGFP	238	Literature benchmark

Key References

1. Pedelacq et al. 2006. Nat Biotechnol 24:79-88 (sfGFP)
2. Close et al. 2015. Proteins 83:1225-1237 (TGP surface engineering)
3. Scott et al. 2018. Sci Rep 8:159 (usGFP core packing)
4. Sarkisyan et al. 2016. Nature 533:397-401 (avGFP fitness landscape)
5. Frenzel et al. 2018 (N39D thermostability)